

4.1 Answer each of the following:

- a. Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2 .
- b. Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.
- c. Suppose that a simple regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2 .

a.

$$SST = \sum (y_i - \bar{y})^2 = \sum y_i^2 - N\bar{y}^2$$
$$\Rightarrow SST = 7825.94 - 20 \cdot (19.21)^2$$
$$= 445.458$$

$$R^2 = \frac{SSR}{SST}$$

$$R^2 = \frac{375.47}{445.458} \approx 0.843$$

b.

$$R^2 = 1 - \frac{SSE}{SST}$$

$$\Rightarrow SSE = (1 - R^2)SST \approx 151.65$$

$$\hat{\sigma}^2 = \frac{SSE}{N - 2}$$

$$\Rightarrow \hat{\sigma}^2 = \frac{151.65}{18} \approx 8.425$$

c.

$$\sum \hat{e}_i^2 = SSE$$

$$\Rightarrow R^2 = 1 - \frac{182.85}{631.63} \approx 0.7105$$